🔥 1️⃣ What is Azure Application Gateway?

Azure Application Gateway is:

> A Layer 7 load balancer for HTTP/HTTPS traffic.

Unlike Load Balancer (Layer 4),

App Gateway understands:

- URL paths
- Host headers
- Cookies
- HTTP headers

🧠 When To Use Application Gateway?

Use it when you need:

- ✅ URL-based routing
- ✅ Multiple websites on same IP
- ✅ SSL offloading
- ✅ Web Application Firewall (WAF)
- ✅ Path-based routing
- ✅ Cookie-based session affinity

🔥 Load Balancer vs Application Gateway

Feature	Load Balancer	App Gateway
Layer	4	7
Protocol	TCP/UDP	HTTP/HTTPS
SSL Termination	❌	✅
URL Routing	❌	✅
WAF	❌	✅

🔥 Core Components of Application Gateway

1️⃣ Frontend IP

2️⃣ Backend Pool

3️⃣ Backend HTTP Settings

4️⃣ Listener

5️⃣ Routing Rule

6️⃣ Health Probe (optional custom)

7️⃣ WAF (optional)

🔥 Architecture Example

</> Code

```
Internet
    ↓
Application Gateway (Public IP)
    ↓
Backend Pool (VM1, VM2)
```

🔥 LAB – Basic Application Gateway

🎯 Scenario

2 web servers:

VM1 → `/app1`

VM2 → `/app2`

User access:

```
</> Code

http://PublicIP/app1
http://PublicIP/app2
```

Traffic routed based on URL path.

STEP 1 – Create 2 Web VMs

Install IIS on both.

VM1:

```
Code

App1 page
```

VM2:

```
Code

App2 page
```

STEP 2 – Create Application Gateway

Portal → Application Gateway → Create

Choose:

- Tier: Standard v2
- Public IP
- Dedicated subnet (Required)

⚠️ Important:

Application Gateway must be in its own subnet.

Example:

```
AppGatewaySubnet (/27 minimum recommended)
```

STEP 3 – Backend Pool

Add:

- VM1
- VM2

STEP 4 – Backend HTTP Settings

Configure:

- Protocol: HTTP
- Port: 80

STEP 5 – Create Listener

Define:

- Frontend IP
- Port 80
- HTTP

STEP 6 – Create Routing Rule

Choose:

👉 Path-based routing

Add:

```
</> Code

/app1/* → VM1
/app2/* → VM2
```

Save.

🧪 Test

Open browser:

```
Code

http://PublicIP/app1
```

Shows VM1

```
Code

http://PublicIP/app2
```

Shows VM2